

Cloud CRM System - Integration Architecture & API Contracts

Confidential - Internal Use Only

Version 1.0

1. System Context

The CRM platform integrates with email providers, calendar services, communication tools, marketing platforms, and data enrichment services to provide a unified sales workflow.

External Services: Gmail API (email sync), Microsoft Graph API (Outlook/Calendar sync), Clearbit Enrichment API (company/contact data), Slack API (notifications), DocuSign eSign API (contract signing), Twilio (SMS/voice), Stripe (billing), Zapier (workflow automation).

Internal Services: Lead Service (capture/scoring/assignment), Contact Service (CRUD/enrichment/dedup), Deal Service (pipeline/forecasting), Activity Service (calls/emails/meetings), Report Service (analytics/custom reports), Integration Service (sync orchestration), Notification Service (email/push/in-app), Search Service (Elasticsearch).

2. Email Integration APIs

REQ-CRM-064: Gmail API OAuth 2.0 integration: Full access to read/send/label emails.

Incremental sync using historyId. Push notifications via Cloud Pub/Sub for real-time updates.

REQ-CRM-065: Microsoft Graph API: Mail.ReadWrite and Calendars.ReadWrite scopes. Delta sync for incremental email retrieval. Webhook subscriptions for mailbox changes.

REQ-CRM-066: Email matching engine: incoming emails matched to CRM contacts by From/To/CC email addresses. Fuzzy matching for email aliases. Unmatched emails queued for manual review.

REQ-CRM-067: Email tracking: invisible pixel (1x1 transparent PNG) inserted for open tracking.

Click tracking via URL redirect through CRM tracking domain. GDPR consent check before enabling tracking.

3. Data Enrichment APIs

REQ-CRM-068: Clearbit Enrichment API: auto-enrich contacts and companies on create/update.

Fields: company size, industry, revenue, tech stack, social profiles. Rate limit: 600 requests/minute.

REQ-CRM-069: Clearbit Reveal API: identify anonymous website visitors by IP to company. Match to existing accounts. Create lead if company not in CRM.

REQ-CRM-070: LinkedIn Sales Navigator API: import LinkedIn profile data, InMail integration, mutual connections data.

4. Communication APIs

REQ-CRM-071: Twilio Programmable Voice: click-to-call from CRM, automatic call recording (with consent prompt), call duration tracking, voicemail drop.

REQ-CRM-072: Slack Web API and Events API: post deal alerts to channels, support slash commands (/crm deal <name>), interactive message buttons for quick actions (approve, snooze, reassign).

REQ-CRM-073: DocuSign eSign API: generate contracts from CRM templates, send for signature, track signing status, auto-update deal stage on completion.

5. Sequence Flow: Lead Capture to Assignment

1. Web Form -> Lead Service: POST /api/leads {name, email, company, source, form_data}

2. Lead Service -> Contact Service: GET /api/contacts/search?email={email} (deduplication check)

3. If no duplicate -> Lead Service: create lead record, generate lead_id

4. Lead Service -> Clearbit API: GET /v2/companies/find?domain={email_domain} (enrichment)

5. Lead Service -> Scoring Engine: Calculate lead score from enrichment + engagement data
6. Lead Service -> Assignment Engine: Evaluate territory rules, round-robin within territory
7. Assignment Engine -> Lead Service: Update lead.assigned_to = rep_id
8. Lead Service -> Notification Service: Send assignment notification (email + Slack DM + push)
9. Lead Service -> Activity Service: Create 'Lead Assigned' activity entry
10. Notification Service -> Slack API: Post to #sales-leads channel with lead card (name, company, score, source)